KENIL GANDHI

Singapore | +65 8313 3382 | e1714042@u.nus.edu | linkedin.com/in/kenil-gandhi-532083220 | github.com/ken2k7

COMPUTER SCIENCE UNDERGRADUATE | FULL-STACK AND APPLIED AI

Year 2 Computer Science student at NUS who builds full-stack and applied-AI products: LLM document extraction at KGiSL, a constraint-optimisation planner at the Nebius × NVIDIA Global AI Hackathon, and MediGraph, a health-tech platform I founded.

EDUCATION

National University of Singapore, Singapore **Expected 05/2029**
Bachelor of Computing in Computer Science
Relevant coursework: CS2103T Software Engineering, CS2100 Computer Organisation, CS2040S Data Structures and Algorithms, CS2030S Programming Methodology II (OOP and functional), CS1231S Discrete Structures, MA1522 Linear Algebra
CS2103T individual project: Kenbot, a Java/JavaFX chatbot with 193 JUnit tests and cross-platform CI (github.com/ken2k7/ip)
Online courses: Harvard CS50x: Introduction to Computer Science; CS50's Introduction to Artificial Intelligence with Python
Quantitative finance (self-study, Paul Wilmott Introduces Quantitative Finance): binomial and Black-Scholes option pricing, stochastic asset models, hedging and the Greeks, finite-difference and Monte Carlo methods

SKILLS

Languages: TypeScript | JavaScript | Python | Java | SQL
Frameworks and Tools: React | Vite | Node.js | Hono | FastAPI | Pydantic | SQLite | Leaflet | Git
AI and Optimisation: Claude API | Gemini API | NVIDIA Nemotron | Google OR-Tools (CP-SAT) | LLM document extraction
Engineering: REST APIs | Graph data modelling | Constraint optimisation | Automated testing | Web accessibility (WCAG AA)

WORK EXPERIENCE

KGiSL, Remote | *Software Engineering Intern, AI and Data* **05/2026 – 07/2026**

- Built a Python and SQLite test harness to evaluate Claude API extraction on unstructured motor insurance claims, and documented the practical constraints of intelligent document processing for the team.
- Integrated the OpenWeatherMap REST API into a Python weather-risk ingestion layer in Marvel.ai's microservices.
- Delivered three interactive views for an internal research dashboard in React, Vite and TypeScript, including a custom CSS Grid timeline that prevented layout drift without relying on DOM measurements.

PROJECTS

MediGraph | *Founder and Full-Stack Engineer* **08/2026 – Present**
Health-tech platform that reconstructs a source-traceable patient history from prescriptions and clinical documents. Stack: TypeScript, React, Node.js, Hono, SQLite, Claude and Gemini APIs.

- Designed a graph data model with 11 node types and 10 relationship types, letting caregivers open any condition and review linked visits, medication changes, tests and the source document behind each fact.
- Built LLM document extraction for prescriptions and discharge summaries (Claude or Gemini) that processes a file in about 20 seconds and keeps a character-level link from every extracted fact to its source text.
- Reviewed the data model with a practising physician, adding inpatient admissions and BP, HbA1c and creatinine trends.
- Set clinical safety limits (no inferred medication stops, lab interpretation or untraceable statements), routing uncertain items to caregiver review; added duplicate detection, 35 automated checks and a WCAG AA accessibility audit.

InfraSync: AI-Assisted Infrastructure Sequencing Planner | *Nebius × NVIDIA Global AI Hackathon (team of 2)* **10/2026**
Schedules a city's road, water, drainage and fibre works so each street is dug once for several jobs. Stack: Python, FastAPI, OR-Tools (CP-SAT), Pydantic, NVIDIA Nemotron (Nebius Token Factory), React, TypeScript, Leaflet.

- Built the scheduler as a CP-SAT model with 6 hard constraints (task ordering, crew capacity, pavement-cut moratoria, funding windows, corridor traffic limits, engineering compatibility) that decides when a shared trench saves money.
- Designed a FastAPI backend with a Pydantic-validated data model and endpoints for planning, disruption replanning and human approve/reject; unsolvable scenarios fall back to the baseline plan instead of failing.
- Kept every scheduling decision in the deterministic solver, used NVIDIA Nemotron to explain plans in plain English from the solver's actual numbers, and required human approval for each change.
- On a synthetic 11-project, 6-street district, found $115K in trench-sharing savings and 12 fewer road-closure days, and correctly rejected trap cases (low-saving and incompatible pairings).

LEADERSHIP AND ACTIVITIES

NUSSU Student Life Council | *Publicity Executive*: produced visual materials for the NUS Student Life Fair **01/2026 – 05/2026**
Model United Nations | *Delegate*, 3 conferences incl. GCIMUN New York; **Toastmasters, Qatar** | *Treasurer* **2022 – 2024**

AWARDS AND ACCOLADES

2025 | 1st of 76 teams, as team lead | Carnegie Mellon University Qatar Pi Day
2021 | Semifinalist, IIT Bombay E-Cell Eureka Junior startup competition | 7th of 150+ speakers, Qatar Schools Debate League
Academics | JEE Advanced All India Rank 6,351 | CBSE Class XII school topper (97.6%)